Reading and Writing

Module 1 · 27 questions 32 minutes

1.

Could field mustard take root on Mars? Could peas? You might guess the answer is flatly no, and yet a Dutch research team recently demonstrated that seeds of many ordinary plant species will germinate in soil engineered to imitate Martian conditions, so long as they are watered. A handful of species even fared *better* in the imitation Martian dirt than in soil from home: field mustard seeds germinated at a 30 percent rate in the artificial ground, against a mere 4 percent of those planted in ordinary terrestrial soil.

Which choice best states the main purpose of the text?

To resolve open questions about reproducing the conditions of Martian soil in a lab on Earth
To share an unexpected result concerning plants raised in dirt designed to mimic that of Mars
To examine a startling outcome regarding water's part in growing plants in fake Martian dirt
To lay out a sweeping study of how the makeup of Martian and Earth soils differs

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2.

The following text is from a work of original fiction. Reva and her husband, Anil, are flying from the United States to India.

Anil lays his hand on hers while the plane begins to gather speed. Then Boston tips away beneath them and they rise without effort above a blackened Atlantic.

As used in the text, what does the word "gather" most nearly mean?

Demand
Gain
Doubt
Accept

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3.

With her 2019 debut *Saltmarsh*, the novelist Wren Adakai proved herself one of the boldest fantasy writers around today. By placing a Diné hero at the story's heart, as Adakai herself is Diné, the book also helped open a door for Indigenous people inside the fantasy genre. Yet she is far from the only Indigenous author now reshaping fantasy: Marcus Ihaka, of the Māori people of Aotearoa New Zealand, has taken up the form as well. His much-praised 2022 novel *Tide-Singer* is set within a Māori coastal town.

Which choice best describes the function of the underlined sentence?

It forecasts that authors who are not Indigenous will write more Indigenous characters into their stories.

It disputes the common belief that Wren Adakai drew *Saltmarsh* from events in her own life.

It underscores just why the depiction of Indigenous people in the novel *Saltmarsh* matters so much.

It questions why so many readers with no love of fantasy went out and bought *Saltmarsh*.

4.

On a well-known board-gaming site, the game Terraforming Mars holds an average user score of 8.4 out of 10. When only a few people have chimed in, an average can deceive; here, though, more than 90,000 separate users have now rated the game—an _____ number for one to trust that its average score holds up.

Which choice completes the text with the most logical and precise word or phrase?

excessive

adequate

accurate

insufficient

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5.

Community science sets professional scientists alongside amateur enthusiasts to study a subject, and often makes research both effective and absorbing. It can draw people into conservation work, kindle young interest in science, and enlarge the data researchers gather. The approach was what carried a study by the biologist Ileana Sandoval and colleagues on how the colour of a butterfly bears on its choice of flower, a study built from findings sent in by hundreds of local students and community members across southern Manitoba.

Which choice best describes the overall structure of the text?

It opens with a study's subject, sets out why that study matters, and then presents its results.

It names an approach to research, sets down several benefits it carries, and then points to a study that put the approach to use.

It urges a new approach to research, notes how the public regards that approach, and then describes one application.

It traces how a kind of collaboration developed, shows it at work within a field, and then proposes further projects.

6.

Genetic material passing from a parent to its offspring is vertical gene transfer; genetic material exchanged between organisms standing in no parent-offspring relation is horizontal gene transfer. Horizontal transfer is common in prokaryotes, the single-celled organisms that include bacteria such as *Brevibacillus borstelensis* and *Lactococcus lactis*, whereas in eukaryotes—that is, multicellular organisms—it has seldom been observed. New studies suggest, however, that horizontal gene transfer runs more common in eukaryotes than anyone first thought.

Which choice best states the function of the underlined sentence in the text as a whole?

- It draws a distinction between how two biological processes work.
- It contrasts how frequently a biological phenomenon has turned up in two different classes of organism.
- It explains why a widespread view of a biological process is mistaken.
- It claims that the two biological phenomena are more alike than they may at first seem.

7.

It is no guarantee of a book's commercial success, yet _____ can matter greatly to that success—a shrewd marketing push helped turn Richard Bachman's novel *The Regulators* into the fifth-best-selling novel of 1996.

Which choice completes the text with the most logical and precise word or phrase?

- originality
- publicity
- complexity
- practicality

8.

That work by Ellen R. McGrattan—a University of Minnesota economist whose subject is financial policy—turns up so often in other scholars' citations _____ how useful her research proves to her peers: fellow economists plainly find her studies valuable for scholarship of their own.

Which choice completes the text with the most logical and precise word or phrase?

- forestalls
- belies
- underscores
- overshadows

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9.

Mean Adhesion Strength at Four Voltages

Voltage (V)	Mean adhesion strength (kPa)
0	7
1	7
2	16
3	38

Naledi Motaung and colleagues showed that a low direct-current electrical field laid across graphite — a conductor — together with a gel of acrylamide (AAm) can make materials cling to one another more strongly. At certain voltages the adhesion strength — the stress in kilopascals (kPa) needed to pull the materials apart — ran high, above 30 kPa. Applying a direct-current field of positive voltage does not on its own bring stronger adhesion, though, as is clear from the fact that _____

Which choice most effectively uses data from the graph to complete the statement?

mean adhesion strength stood at 0 kPa when no voltage was applied.
mean adhesion strength peaked at 3 V, reaching roughly 30 kPa.
mean adhesion strength at 2 V fell below what it was at 1 V and at 3 V alike.
mean adhesion strength at 1 V came out at roughly what it was with no voltage at all.

10.

Total Areas of Five Wildlife Refuges in the United States

Wildlife refuge	State	Area (square miles)
Cedar Marsh Refuge	Oregon	4,780
Blue Heron Refuge	Michigan	3,510
Willow Bend Refuge	Iowa	1,240
Salt Flat Refuge	Nevada	2,090
Pine Ridge Refuge	Georgia	4,905

Pine Ridge Refuge ranks among the largest wildlife refuges anywhere in the country, spanning 4,905 square miles across present-day Georgia. Willow Bend Refuge, over in Iowa, is far more modest, taking up a mere _____

Which choice most effectively uses data from the table to complete the comparison?

1,240 square miles.
2,090 square miles.
4,780 square miles.
3,510 square miles.

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11.

The American foxhound descends, as most dog breeds of the Americas do, from dogs carried over from Europe in the colonial era. The Xoloitzcuintli, by contrast, belongs to the handful of breeds descended from non-European populations that were here before colonization. Researchers added a further breed to that handful in 2013: the Carolina dog, native to rural pockets of the southeastern United States, which carries a distinctive genetic tie to East Asian populations, themselves the forebears of the precolonial dogs of the Americas.

What is the subject of the main distinction the text draws between the dogo argentino and the Xoloitzcuintli?

- The size, the colouring and the other traits the two breeds are typically associated with
- Where the populations that the two breeds descend from came from
- How much the two breeds gave to the ancestry of more recently developed breeds
- How quickly each breed spread out of its place of origin into the southeastern United States

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12.

The verse below is adapted from "A Remembrance of Puerto Cabello—The Mangle," a poem D. A. Lozano wrote in 1844 and Agnes Blake Poor translated in 1918. Puerto Cabello is a town on Venezuela's northern coast.

And while outside the tempest is raving o'er the ocean, And the ship is madly driving on some lone and desert shore; Thy warm and land-locked waters swell with an easy motion, And gently glides the light pirogue [canoe] at dipping of the oar.

Which choice best states the main purpose of the text?

- To stress how fierce a storm is that is expected to reach Puerto Cabello shortly
- To convey how the speaker regards ordinary daily life in Puerto Cabello
- To weigh the materials from which two sorts of boat are built
- To set the calm of the waters at Puerto Cabello against the violence of waters beyond them

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13.

Biologists Hana Suzuki and Marcus Bell carefully reviewed a body of studies of animals such as zebra finches and degus, creatures that have been observed both under laboratory conditions and out in the wild, in order to check whether the findings differed between the two settings. They found, for one example, that among degus the wild individuals were most active during the daytime hours while the captive ones were not. Suzuki and Bell therefore concluded that the laboratory environment was in all likelihood _____

Which choice most logically completes the text?

- better than wild settings as a place to study degus' patterns of rest and activity.
- affecting the results for both degus and zebra finches.
- better suited to studying zebra finches than to studying degus.
- interfering with degus' patterns of rest and activity.

14.

Could a field mustard plant grow on Mars? Could a pea plant? You might assume the answer is plainly no, yet a Dutch research team recently showed that seeds from many common plant species will readily sprout in soil built to mimic Martian conditions, so long as they are given water. In fact, a few species fared *better* in the mock Martian soil than in ordinary Earth soil: field mustard seeds sprouted at a 30 percent rate in the simulated Martian dirt, versus only 4 percent of those sown in soil from their own home planet.

According to the text, what percentage of field mustard seeds planted in Martian soil sprouted?

18 percent
30 percent
100 percent
5 percent

15.

The following text is adapted from Daniel Defoe's 1704 nonfiction book *The Storm*.

If I judge right, 'tis the duty of an historian to set everything in its own light, and to convey matter of fact upon its legitimate authority, and no other... that where a story is vouched to him with sufficient authority, he ought to give the world the special testimonial of its proper voucher, or else he is not just to the story: and where it comes without such sufficient authority, he ought to say so; otherwise he is not just to himself.

Which choice best states the main idea of the text?

A historian seldom has every piece of information bearing on an event, so no one work of history should be taken as final.
A historian ought to quote none but those figures whom the world at large holds credible.
A historian ought to make plain how far each of the sources drawn on can be trusted.
Because a historian cannot tell whose memory to rely on, eyewitness accounts of events are best avoided.

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16.

Many writings by Sallust, a Roman historian of the 1st century BCE—among them his account of the Roman Republic, the *Histories*—have not survived. A reader _____ to take in one of Sallust's surviving works, not one known merely through later writers' references, might instead track down his *Bellum catalinae*, which chronicles an attempted coup.

Which choice completes the text so that it conforms to the conventions of Standard English?

hoped
hoping
have hoped
hope

17.

In English metrical verse, trisyllabic words _____ as dactyls—"devious" and "particle" among them—carry one stressed syllable trailed by two unstressed ones.

Which choice completes the text so that it conforms to the conventions of Standard English?

classified
can be classified
are classified
have been classified

18.

Lengthy narratives dwelling on the heroic or the divine are what mark out epic poetry as a genre. *The Argonautica*, a Greek epic poem, goes back to the _____ its heroic narrative holds modern readers still.

Which choice completes the text so that it conforms to the conventions of Standard English?

3rd century BCE,
3rd century BCE yet,
3rd century BCE, yet
3rd century BCE.

19.

In 2019, Néstor Ovalle _____ as a marine biologist at the University of Otago, studying octopuses.

Which choice completes the text so that it conforms to the conventions of Standard English?

working
was working
to work
to have worked

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20.

Emulsifiers, which turn up in products from food to cleaners, are compounds that hold an emulsion together—an emulsion being two or more liquids mixed together that would not otherwise blend easily. Emulsifiers of the sort _____ reach for, oleic acid among them, let these scientists work oil together with water into homogeneous formulations, lotions and perfumes among them.

Which choice completes the text so that it conforms to the conventions of Standard English?

cosmetic chemists,
cosmetic chemist's,
cosmetic's chemists,
cosmetic chemists',

21.

As a leading figure in the American Anti-Slavery Society during the mid-1800s, Charles L. Reason of _____ a central role in the movement to abolish slavery in the United States.

Which choice completes the text so that it conforms to the conventions of Standard English?

Kentucky. Played
Kentucky: played
Kentucky; played
Kentucky played

22.

When trying to reconstruct a glacier's recent history, glaciologists like Dr. Priya Anand at the University of Alberta rely in part on data about the glacier's physical dimensions and its rate of _____ the Athabasca Glacier in Alberta, Canada, the Canadian Ice Service gathers such data in several ways: by trading information with provincial monitoring programs, by pairing maps of the glacier with detailed field descriptions of it, and by keeping a database of measurements such as the glacier's length (6 km).

Which choice completes the text so that it conforms to the conventions of Standard English?

features and for
features. For
features for
features, for

23.

While researching a topic, a student has taken the following notes:

- A dinosaur fossil specimen dating from the Late Cretaceous period has been given the nickname Sue.
- That geological period wrapped up over 65 million years in the past.
- Sue belongs to the genus *Tyrannosaurus*.
- The specimen is kept on display at the Field Museum of Natural History.
- That museum stands in Chicago, Illinois.

The student wants to identify the institution in which Sue is housed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Sue is housed at the Field Museum of Natural History in Chicago, Illinois.
Sue is a *Tyrannosaurus* fossil specimen that has been given a nickname.
A *Tyrannosaurus* specimen nicknamed Sue can be found in Chicago, Illinois.
Dating to the Late Cretaceous, the *Tyrannosaurus* specimen Sue lived over 65 million years ago.

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24.

While researching a topic, a student has taken the following notes:

• Lake Baikal is located in the mountains of southern Siberia, in Russia.
• It reaches a maximum depth of about 5,387 feet.
• Crater Lake is located in the Cascade Range, in the state of Oregon.
• It reaches a maximum depth of about 1,949 feet.
• These two bodies of water are among the few dozen lakes on Earth classified as "ancient lakes."
• An ancient lake is one that has held water continuously for more than a million years.

The student wants to emphasize the similarity between the two mountain ranges. Which choice most effectively uses relevant information from the notes to accomplish this goal?

There are a number of ancient lakes located on different continents around the world.

Lake Baikal and Crater Lake are both "ancient lakes," each having held water continuously for more than a million years.

Lake Baikal reaches a depth of 5,387 feet, whereas Crater Lake reaches a depth of 1,949 feet.

Though both are ancient lakes, Lake Baikal and Crater Lake lie on different continents.

25.

While researching a topic, a student has taken the following notes:

- The Japanese American artist Isami Doi made the 1935 color relief print *Sheepshead Bay*.
- It depicts boats as they near a dock.
- Both intaglio and relief are printmaking methods in which an image is cut into a printing block, inked, and pressed onto paper.
- In intaglio printing, ink goes into the grooved, carved parts of the block.
- In relief printing, ink goes onto the raised, uncarved parts of the block.

The student wants to contrast intaglio and relief printing. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Relief printing is the reverse of intaglio: the ink goes not into the block's carved grooves but onto its raised, uncarved parts.

Relief printing, not intaglio, is the method Doi used to depict boats nearing a dock in the 1935 print *Sheepshead Bay*.

Unlike relief printing, in which ink goes on the uncarved parts of the block, intaglio uses the block to press the image onto paper.

In both relief and intaglio, an image is cut into a printing block, leaving grooves that ink is applied to.

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26.

While researching, a student has taken the following notes:

• Rosalina Vega is a Peruvian American artist who builds sculptures out of LED lights.
• *Reed Island* (2011), from her Reed series, is made of smooth multicoloured LED domes.
• *Golden Thread* (2013), from her Fisher series, is made of tangled blue and white LED tubes.

The student wants to emphasize a difference between the two sculptures. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Vega turns often to LED lights, and yet the smooth LED domes of *Reed Island* stand in contrast to the tangled LED tubes of *Golden Thread*.

Reed Island and *Golden Thread* are two LED sculptures by Vega.

Vega brought out *Reed Island*, part of her Reed series, in 2011.

Plenty of sculptures, *Reed Island* and *Golden Thread* among them, work LED lights into domes or tubes.

27.

While researching a topic, a student has taken the following notes:

- The Clovis First hypothesis holds that an ancient people called the Clovis were the first to settle North America.
- The Clovis are thought to have arrived about 13,000 years ago.
- Stone tools from the Saltville site in Virginia date to roughly 16,000 years ago.
- Those tools support the idea that people were living on the continent before the Clovis arrived.
- Over a dozen sites across North America have since yielded pre-Clovis material.

The student wants to explain the implications of the discovery at the Saltville site. Which choice most effectively uses relevant information from the notes to accomplish this goal?

According to the Clovis First hypothesis, the Clovis, an ancient people, were the earliest to settle North America, showing up about 13,000 years ago.

Saltville, in Virginia, ranks among the dozen-plus North American sites that have produced pre-Clovis material.

The roughly 16,000-year-old stone tools from Saltville, a site in Virginia, deserve attention: they bear importantly on the Clovis First hypothesis.

The roughly 16,000-year-old stone tools unearthed at Saltville cut against the Clovis First hypothesis, hinting that people reached North America before the Clovis did.

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Reading and Writing

Module 2 · 27 questions 32 minutes

1.

Large electronic collections of stored text called corpora—the Oxford English Corpus among them—can be used to gauge how often _____ words occur. (Words that appear mainly in specialized or academic writing, by contrast, tend to be underrepresented in them.) Such a corpus might show, for instance, that “way” is the fourth most common noun in standard English.

Which choice completes the text with the most logical and precise word or phrase?

accidental
conventional
trivial
profound

2.

Political scientists report a curious gap: voters say they favor candidates who take nuanced stances and signal a readiness to compromise, but when handed speeches voicing such stances alongside speeches voicing _____ ones, those same voters end up rating the unbending rhetoric of the latter more highly.

Which choice completes the text with the most logical and precise word or phrase?

dogmatic
vacillating
banal
disingenuous

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3.

Phosphine (PH_3) — on Earth a gas arising almost only from biological sources — was apparently detected in the cloud deck of Venus in 2020, and the finding stays controversial for all the prospect of independent confirmation, partly because Venus is taken to be uninhabitable. Weighing its plausibility, William Bains et al. modelled abiotic routes to PH_3 — geochemical, atmospheric, photochemical — and none accounts for the levels observed. Venusian PH_3, if real, would point to gaps in the present consensus on the chemistry of Venus.

Which choice best describes the overall structure of the text?

It sets the levels of one gas on Venus against those on Earth, sketches the chemistry accounting for the difference, and then takes up some practical difficulties in studying the gas on Venus more closely.

It explains why the consensus view of one gas in the atmosphere of Venus has lately grown controversial, expands on a team's grounds for doubting that consensus, and then suggests that further observation will probably be needed to settle the matter.

It outlines recent attempts to confirm that a gas is present in the atmosphere of Venus, sums up a team's assessment of where those attempts fell short methodologically, and then explains why the team doubts the gas will be detected.

It sets out a consensus long held about a gas in the atmosphere of Venus, describes an experiment that overturned it, and then predicts the gas will shortly be detected directly.

4.

A team of microbiologists took surface temperatures from a range of wild mushrooms, *Amanita muscaria* and a species of the genus *Thelephora* among them. Readings came in between 1.4 and 5.9°C below the temperature of the surrounding air, a gap the team _____ to water evaporating off the mushrooms and cooling them as it went.

Which choice completes the text with the most logical and precise word or phrase?

conceded

prescribed

attributed

amended

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5.

Diadromous fish move between freshwater and marine biomes over their life cycle. Because this migration is obligate, diadromous fish can be _____ ones that are only euryhaline (that is, able to withstand high salinity): the euryhaline blackchin tilapia tolerates high salinity, but its life cycle involves no move to a different biome, unlike that of the diadromous wild salmon.

Which choice completes the text with the most logical and precise word or phrase?

demarcated from

derived from

conflated with

reconstituted as

6.

Tunica, an Indigenous language of the lower Mississippi Valley in what are now the United States, has *mili* for "red," while *milimilita* is what one says of several red things. The phenomenon at work—a piece of a root word coming back, sometimes altered, inside another word tied to that same root—is called reduplication. Here the whole word *mili* returns inside *milimilita*. Reduplication of just this kind turns up again and again throughout the Tunica language.

Which choice best describes the overall structure of the text?

It sets out particular Tunica words, describes the general linguistic phenomenon they exemplify, and then states that the phenomenon is frequent in Tunica.

It picks out the words occurring most often in Tunica, explains what makes them hard to render in English, and then offers other languages they can be rendered into.

It explains the phenomenon of reduplication, reviews what has made it controversial among scholars, and then argues that Tunica could settle the controversy.

It describes how Tunica stands to several other languages, raises a question about that relation, and then answers it.

7.

For all that its _____ was claimed, a good deal of the income-inequality analysis carried out through the 2010s kept a myopic focus on a handful of countries across North America and Western Europe, limited data availability being part of the reason. Researchers would put that shortcoming right later, once fresh access opened to administrative records held in African nations such as Ivory Coast and Eastern European ones such as Poland.

Which choice completes the text with the most logical and precise word or phrase?

novelty

immutability

exhaustiveness

pervasiveness

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8.

Like every other river delta, the delta the Kura River builds in Azerbaijan is _____: an ever-shifting web of channels and tongues of land that alter in size and in shape as the river lays down fresh sedimentary particles where its water runs into the Caspian Sea.

Which choice completes the text with the most logical and precise word or phrase?

sustainable

unrivaled

immutable

dynamic

9.

Optimal foraging theory (OFT) proposes that how animals forage reflects cost-benefit trade-offs that shift from species to species and with changing ecological conditions. One such condition is lunar intensity: Nathan Upham and John Hafner reported it negatively linked to foraging by the desert kangaroo rat, whereas Eduardo Fernández-Duque and colleagues reported it positively linked to foraging by Azara's night monkeys. OFT accounts for the gap—because the monkeys depend more heavily on sight, a brighter moon helps them more than it helps the kangaroo rats.

According to the text, the difference between Upham and Hafner's findings and Fernández-Duque and colleagues' findings can be attributed to which difference between desert kangaroo rats and Azara's night monkeys?

The kangaroo rats cut back their foraging as lunar intensity rises, while the monkeys forage more.

The kangaroo rats face more danger from predators than the monkeys do.

The kangaroo rats depend on sight less than the monkeys do.

The kangaroo rats run into lunar-intensity levels unlike those the monkeys meet.

10.

The following text is from Halvard Reim's 1931 novel *The Long Ascent*.

What we mean to set down here concerning Emil Barrow—and we set it down not on his account (any reader will soon enough take him for the plain, agreeable young man he is) but on account of the thing itself, which strikes us as amply deserving to be set down (though it should be said for Barrow that the thing is *his*, and that not every man is handed such a thing to carry)—belongs to an age well behind us, has acquired, one might say, the dull sheen that time lays over what it keeps, and asks of necessity to be put in the tenses reserved for the remotest past.

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What does the text most strongly suggest about the story of Emil Barrow?

Although any life, however dull its owner, makes an interesting story because no two people are alike, the passage of time makes it hard to see just where this story's interest lies.

That the story concerns a person of no special consequence matters less than its age and the manner of telling it therefore requires, both of which mark the story out as important.

An unremarkable person has had a remarkable thing happen to him, though a case could be made that some of what makes the story valuable rubs off on the man at its center.

As with every story about a life of no consequence, this one has to be told in a certain manner if the audience is to see why it matters at all.

11.

More than 600 languages besides English are spoken in New York City: Bambara can be heard in the Kingsbridge neighborhood, Ghale in Jackson Heights. Speakers of Chinese languages live for the most part in two neighborhoods. One is Flushing, which lies within the borough of Queens; Mandarin predominates there. The other is Chinatown, over in Manhattan, where the languages that predominate are Cantonese and Fuzhounese. Mandarin is the language spoken widely across the north of China, while Cantonese and Fuzhounese are the languages spoken widely down in the south. It follows that _____

Which choice most logically completes the text?

Chinese immigrants who reached New York City long ago speak several Chinese languages more often than immigrants who have arrived recently do.

those who emigrate from the north of China settle in Queens as a rule, while those who emigrate from the south settle in Manhattan.

Cantonese and Fuzhounese speakers together outnumber Mandarin speakers among the Chinese immigrants of New York City.

Chinese immigrants move house between Queens and Manhattan with some regularity once they arrive, rather than remaining in one borough.

12.

Copenhagen sees heavy foot traffic, but merely copying one feature linked to its walkability—say, its dense network of narrow lanes—may not be enough to get people walking more in other cities. As the urbanist Deval Rao contends, our grasp of how individuals decide whether to walk remains too shaky: some studies stress demographics, others the reach of public transit, and so on, yet walking choices emerge from tangled settings in which many conditions and needs shape what a person does.

Based on the text, Rao would most likely agree with which statement about studies of decision-making about walking?

They would improve if researchers pinned down what cities with heavy foot traffic have in common.

They have exaggerated the extent to which people's walking decisions differ from one another.

They tend to mislead because they overlook the single most important influence on walking.

They are unlikely to turn up solid proof that any one factor reliably predicts whether people walk.

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13.

Hydrogen has many uses, from the making of steel to heating a home. Producing it is carbon intensive as things stand, but Priya Raghunathan is one of many scholars asking whether hydrogen can be made at scale by electrolysis, which emits no carbon where renewable energy drives it. Obstacles may keep electrolytic hydrogen from taking over energy systems, Raghunathan suggests, yet there is use in what it can do to decarbonise steel and heating, industries where cutting carbon emissions is particularly hard.

Which statement about the use of hydrogen in the steel and heating industries is most strongly supported by the text?

Applications for electrolytic hydrogen are less far along in these industries than in most other carbon-intensive ones.

That electrolytic hydrogen cannot yet be produced at large scale has damped these industries' interest in hydrogen-based applications.

Worries about how carbon intensive its production is have held these industries back from taking up electrolytic hydrogen.

Part at least of what electrolytic hydrogen could do for these industries' emissions is still unrealised.

14.

Kinds of New Year Resolution Made, by Age of the Resolution-Maker

Resolution type	18–29	30–49	50–64	65 and older
Health and exercise	77	81	78	75
Finances	66	61	54	45
Personal relationships	61	52	57	50
Hobbies	64	52	50	44

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A survey run in January 2024 by the Larchmont Research Group found that three US adults in ten set at least one New Year resolution, and that half of those who set one go on to set more than a single one. The table sets out, by age bracket, the share of the resolution-makers who make each particular kind of resolution.

Which choice best presents a conclusion about the habits of New Year's resolution makers that is best supported by information in the text and the table?

Most US adults who make resolutions about health and exercise also make resolutions in several further categories.

Compared with resolution-makers who fall in the 30-to-49 age bracket, those in the 50-to-64 bracket are likelier to make resolutions about their personal relationships, and at the same time less likely to make resolutions about their finances.

Across US adults, the likelihood of making a New Year resolution falls with age whatever the type of resolution.

Resolution-makers aged 18 to 29 are likelier than those aged 30 to 49 to make resolutions about health and exercise.

15.

Among highly social mammals, the company of others of the same species has been seen to blunt stress. To find out whether this so-called social buffering runs in reptiles too, researchers under Iona Brackenridge tracked how wild southern Pacific rattlesnakes (*Crotalus helleri*) answer stress across three treatments: left alone, given a rope, and given a fellow snake for company. What the team compared was the percentage change from baseline heart rate to peak heart rate in answer to a harmless disturbance, a higher figure marking greater stress.

Which finding, if true, would most directly support the idea that social buffering occurs among *C. helleri*?

Peak heart rates on average ran highest in the snakes left alone, though companioned snakes and rope snakes did not differ in average peak heart rate.

Companioned snakes showed both a lower average baseline heart rate and a lower average peak heart rate than snakes left alone or snakes given a rope did.

The average percentage change in heart rate ran lower in companioned snakes than in snakes left alone or snakes given a rope.

Snakes left alone carried higher average baseline heart rates than snakes in the other treatments, and yet the average percentage change in heart rate came out smaller in them than in companioned snakes.

16.

On March 3, 2019, a team of Colorado-based researchers—Hannah Delgado, Ravi Menon, and Sofia Kraus—_____ climate readings from several sites along Peru's Cordillera Blanca. By 9:00 a.m. the air stood at 14°C at RB2, the highest of the sites, and it had drifted up to 14.9°C by 10:30 p.m.

Which choice completes the text so that it conforms to the conventions of Standard English?

compiled,

compiled

compiled;

compiled:

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17.

One might expect the major-party campaigns in a national US election to concentrate on the most populous states. Where polling and past voting records make the result in a given state a foregone conclusion, however, no campaign spends resources there. In the end a state's voting record and polling data, not its population size, _____ how much it matters to campaigns.

Which choice completes the text so that it conforms to the conventions of Standard English?

determines

has determined

determining

determine

18.

The Organisation for Economic Co-operation and Development (OECD) keeps comparative price-list figures for its thirty-eight member countries. In July 2021, for one, a hypothetical basket of goods that in the United States cost 100 US dollars (USD) would have come to 39 USD and to 130 USD in the fellow OECD _____ and Iceland, respectively.

Which choice completes the text so that it conforms to the conventions of Standard English?

nations; Colombia
nations, Colombia,
nations, Colombia
nations Colombia

19.

The liquid hydrocarbons decane ($C_{10}H_{22}$) and octane (C_8H_{18}) are assigned molar masses of 142.28 and 114.23 _____ mass lets chemists translate a given quantity of one reactant into the exact quantity of product it will yield).

Which choice completes the text so that it conforms to the conventions of Standard English?

g/mol (respectively, in stoichiometry, molar
g/mol, respectively, (in stoichiometry, molar
g/mol, respectively (in stoichiometry, molar
g/mol. Respectively, in stoichiometry (molar

20.

Yash Sondhi and Samuel Fabian, both entomologists, have set out to explain why moths fly so erratically around lights after dark. Aware that flying insects hold their backs turned toward the sun during daylight, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

the researchers' theory is that moths mistake nighttime lights for the Sun, continually trying to reorient their bodies while flying near such lights.
the researchers theorize that moths, mistaking nighttime lights for the Sun, continually try to reorient their bodies while flying near such lights.
moths mistake nighttime lights for the Sun and continually try to reorient their bodies while flying near such lights, the researchers theorize.
moths continually try to reorient their bodies while flying near nighttime lights, the researchers theorize, mistaking such lights for the Sun.

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21.

It was the sort of puzzle that would set any music historian's mind into _____ that hidden thread able to tie together figures as unlike one another as the Baroque composer Barbara Strozzi, the ragtime pianist Scott Joplin, and the Estonian minimalist Arvo Pärt.

Which choice completes the text so that it conforms to the conventions of Standard English?

motion, finding:
motion. Finding
motion; finding
motion: finding

22.

The Organisation for Economic Co-operation and Development (OECD) compiles comparative price-level figures across its thirty-eight member states. By those figures, as of July 2021 a “basket” holding goods and services worth 100 US dollars (USD) within the United States would have run to roughly 66 USD or so in fellow OECD member _____

Which choice completes the text so that it conforms to the conventions of Standard English?

nation Latvia.
nation: Latvia.
nation, Latvia.
nation—Latvia.

23.

Hoping to understand what keeps consumers loyal to particular online retailers, Vijoleta Vrhovac and colleagues ran a survey built on 58 statements standing in as indicators of what customers experience when their online purchases arrive. Each statement fell under a topic—number 13, about how eagerly a customer waits for ordered goods to arrive, came under joyful anticipation—and respondents, every one of them from Serbia, a country described as having a developing economy, rated how much each statement mattered to their own experience. Package aesthetics drew low importance from participants while speedy delivery and package tracking drew high importance, yet the researchers warned against extending the findings to customers at large, suggesting that _____

Which choice most logically completes the text?

further research drawing on participants from countries at varying levels of economic development is needed to establish whether delivery time and tracking outweigh package design for customers broadly.
customers in countries whose economies are more developed than Serbia’s are likely to weigh package design relatively more heavily than delivery time and transparency.
the greater weight given to delivery time and tracking over package design might not show up were the survey put to a larger group of Serbian participants.
online retailers working across a variety of countries stand a better chance of winning customer loyalty by delivering faster and offering tracking than by improving their packaging.

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24.

Work on deflecting asteroids has settled on secondary bodies such as S/2015 (190208), a moonlet going round the near-Earth asteroid 2006 AQ. NASA drove a probe deliberately into just such a body in 2022 and altered its orbit successfully. Whether 2006 AQ and other primary bodies would answer the same way, _____ nobody has yet demonstrated.

Which choice completes the text with the most logical transition?

for example,
admittedly,
moreover,
likewise,

25.

While researching a topic, a student has taken the following notes:
• Cinematographers handle cameras and lighting.
• They assist in turning a film director's ideas into visual images.
• Julie Taymor directed *Frida* (2002).
• Rodrigo Prieto served as the film's cinematographer.

Which choice most effectively uses information from the given sentences to provide an example of a film Prieto worked on?

Rodrigo Prieto, a cinematographer, worked alongside the director Julie Taymor.
Rodrigo Prieto served as cinematographer on the 2002 film *Frida*, to give one example.
Directors in filmmaking work alongside others to turn their ideas into visual images that we meet on screen.
Rodrigo Prieto, as cinematographer, handles cameras and lighting so as to turn a director's ideas into visual images.

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26.

Early in the 1900s the sculptor and collector Gertrude Whitney stood among the foremost champions of avant-garde art, helping a good many US avant-garde artists to publicity and exposure. _____ Whitney offered a leading New York museum more than 500 avant-garde works for display, but the museum refused, and so she founded her own: the Whitney Museum of American Art.

Which choice completes the text with the most logical transition?

To this end,
In other words,
In contrast,
Granted,

27.

While researching a topic, a student has taken the following notes:

• Modularity of mind holds that the mind is composed at least partly of innate neural structures (modules) carrying out fast, necessary tasks.

• The cognitive scientist Jerry A. Fodor proposed in 1983 that low-level cognitive systems (perception and language, say) are modular.

• Under Fodorian modularity, high-level systems (reasoning, say) are not modular.

• The cognitive scientist Peter Carruthers put forward the massive modularity hypothesis (MMH) in 2003.

• MMH stretches modularity to take in every cognitive system.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

Carruthers, after Fodor's 1983 hypothesis, proposed that modularity of mind takes in every cognitive system.

Carruthers put forward the massive modularity hypothesis in 2003, dissenting from Fodor's earlier hypothesis that innate neural structures compose the mind.

In treating some though by no means all cognitive systems as modular, Fodorian modularity turns out to define modularity less expansively than MMH does.

Fodor's hypothesis and Carruthers's diverge over whether low-level cognitive systems, perception and language among them, count as modular.

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Reference

Circle: $A = \pi r^2$ · $C = 2\pi r$ · arc/sector proportional to the central angle; a circle has $360° = 2\pi$ radians.

Rectangle: $A = \ell w$ · **Triangle:** $A = \frac{1}{2}bh$ · **Right triangle:** $a^2 + b^2 = c^2$

Special right triangles: 30°-60°-90° sides x, $x\sqrt{3}$, $2x$ · 45°-45°-90° sides s, s, $s\sqrt{2}$

Rectangular solid: $V = \ell wh$ · **Cylinder:** $V = \pi r^2 h$ · **Sphere:** $V = (4/3)\pi r^3$

Cone: $V = (1/3)\pi r^2 h$ · **Pyramid:** $V = (1/3)\ell wh$

The number of degrees of arc in a circle is 360. The sum of the measures of the angles of a triangle is 180.

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Math

Module 1 · 22 questions 35 minutes

1.

An environmental scientist needs to collect at least 38 samples of red clay soil for analysis. If r represents the number of samples the scientist collects for analysis, which inequality represents this situation?

$r \leq 38$

$r - 38 \geq 38$

$r - 38 \leq 38$

$r \geq 38$

2.

If $4x - 28 = -12$, what is the value of $x - 7$?

-25

-12

-4

-3

3.

A teacher is creating an assignment worth 42 points. The assignment will consist of questions worth 1 point and questions worth 2 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 2-point questions?

$3xy = 42$

$3(x + y) = 42$

$2x + y = 42$

$x + 2y = 42$

4.

Square B is a scaled copy of square A. The length of each side of square A is 13 inches, and the scale factor from square A to square B is 3. What is the side length, in inches, of square B?

10

13

16

39

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5.

$y < -4x + 12$
Which point (x, y) is a solution to the given inequality in the xy-plane?

(0, 13)
(−1, 17)
(−4, 0)
(4, −1)

6.

$(5/9)(x - k) = k(x - 9/5)$
In the given equation, k is a constant. The equation has no solution. What is the value of k?

−5/9
−9/5
9/5
5/9

7.

A beekeeper's initial bee-colony population was 1,600 bees, with a goal of 2,975 bees. The population begins at 1,600 and increases by 110 bees per week for the first two weeks, then by 165 bees per week until the goal is reached. At the end of week w (where w > 2), which function gives the predicted number of bees still needed to reach the goal?

p(w) = −110 + 165w
p(w) = 1,485 − 165w
p(w) = 2,865 + 165w
p(w) = 2,975 − 165w

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8.

A club collects $12 per year from each of its x standard members and $24 per year from each of its y premium members. In one year the club collected $3,600 in membership fees. Which equation represents this situation?

24x + 12y = 3,600
12x + 24y = 3,600
36(x + y) = 3,600
12x + 24y = 3,600(2)

9.

The function g is defined by $g(x) = 10^{3x-1}$. What is the value of x when $g(x)$ is equal to 10,000?

$\frac{4}{3}$

$\frac{5}{3}$

3

4

10.

What is the x-intercept of the graph of $y = (45 - x)/(ax + b)$ in the xy-plane, where a and b are positive constants?

(−b/a, 0)

(0, 45/b)

(0, −45)

(45, 0)

11.

The variables x and y have a quadratic relationship. When x = 0, y = 45. When x = 6, y = 81. When x = 15, y = 0. Which equation represents this relationship?

$y = x^2 - 18x + 45$

$y = -x^2 + 12x + 45$

$y = x^2 + 45$

$y = -3x^2 + 48x - 45$

12.

$x^2 - 24x - 8 = 0$
What is the sum of the solutions to the given equation? (This item is drawn from a sibling March 2025 form to fill a page missing from this recording.)

0

4

8

24

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13.

$$f(x) = x^2 - 4x - 780$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

$f(x) = x^2 - 4x + 195$

$f(x) = (x - 2)^2 - 784$

$f(x) = x(x - 4) - 780$

$f(x) = (x + 26)(x - 30)$

14.

The number of bacteria in a growth medium is expected to increase by 170% every 2 hours during a period of observation. The number of bacteria in the growth medium was estimated to be 5,000 when the period of observation began. Which function P gives the expected number of bacteria in this growth medium t hours after the period of observation began?

P(t) = 5,000(1.70)^(t/2)

P(t) = 5,000(1.70)^(2t)

P(t) = 5,000(2.70)^(t/2)

P(t) = 5,000(2.70)^(2t)

15.

How many times does the graph of the equation $y = 9(a/7)^{x+c} - b$ in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 7$ and $b > c$?

Zero

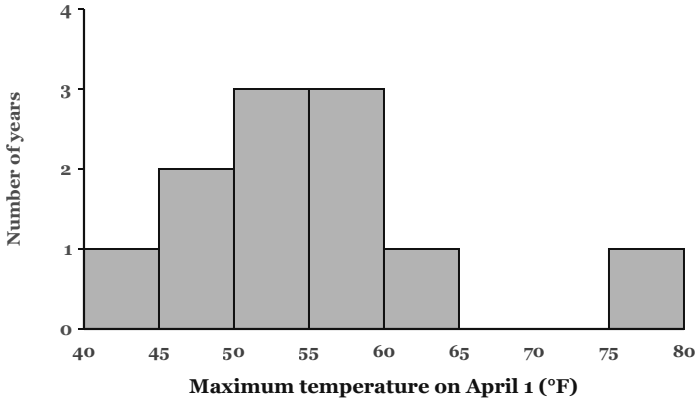
One

Two

Three

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16.



The maximum temperature on April 1, in degrees Fahrenheit (°F), was recorded each year at a certain weather station for 11 years. The histogram summarizes the recorded data set. The temperature of 77.9°F is removed from this data set to create a new data set of 10 temperatures. Which of the following statements must be true?
I. The mean of the new data set is less than the mean of the original data set.
II. The median of the new data set is less than the median of the original data set.

I and II

II only

Neither I nor II

I only

17.

A snack company advertises that bags of pretzels produced by the company weigh, on average, 1 pound each. To test this, Sam selected at random 40 bags of pretzels produced by the company and weighed each bag. Based on his measurements, Sam estimated that for all bags of pretzels produced by the company, the average weight per bag is 1.05 pounds, with a margin of error of 0.06 pounds. For all bags of pretzels produced by the company, which of the following is the most appropriate conclusion about the average weight w, in pounds, per bag?

$0.99 \le w \le 1.05$

$w = 1.05$

$1.05 \le w \le 1.11$

$0.99 \le w \le 1.11$

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18.

A circle in the *xy*-plane has its center at (−4, 2) and the point (−8, 5) lies on the circle. Which equation represents this circle?

$(x - 4)^2 + (y + 2)^2 = 5$

$(x + 4)^2 + (y - 2)^2 = 5$

$(x - 4)^2 + (y + 2)^2 = 25$

$(x + 4)^2 + (y - 2)^2 = 25$

19.

What value of w is the solution to the equation 0.7w − 0.57 = 7(w − 0.001)?

Student-produced response — enter your answer in the grid.

20.

The function $f(t) = 10{,}000(2)^{\frac{t}{850}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

Student-produced response — enter your answer in the grid.

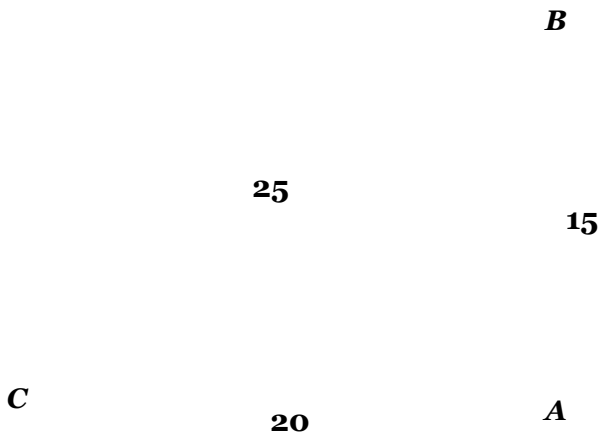
21.

An object's speed is increasing at a rate of 12.60 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

Student-produced response — enter your answer in the grid.

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22.



Note: Figure not drawn to scale.

What is the value of cos *B* in the right triangle shown?

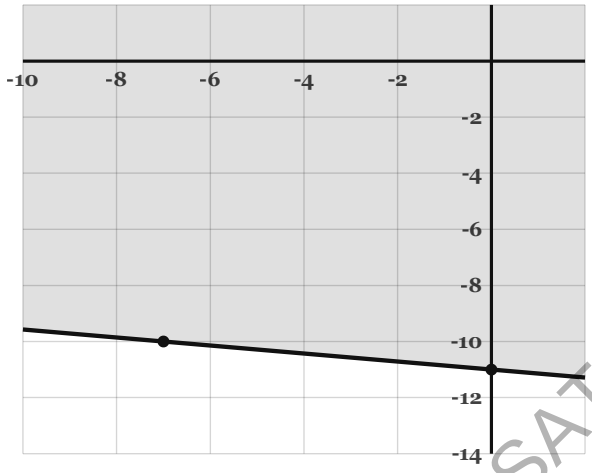
Student-produced response — enter your answer in the grid.

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Math

Module 2 · 22 questions 35 minutes

1.



The shaded region shown represents the solutions to $rx + ty \geq -77$, where r and t are constants. What is the value of r + t?

8

6

−6

−7

2.

x	g(x)
-15	3
-6	0
9	5

The table shows three values of *x* and their corresponding values of g(x), where $g(x) = f(x)/(x + 3)$ and *f* is a linear function. What is the y-intercept of the graph of y = f(x) in the xy-plane?

(0, −6)

(0, 4)

(0, 8)

(0, 24)

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3.

$$(1/9)x - (3r/7)y = 34$$

One of the equations in a system of two linear equations is given, where *r* is a nonzero constant. The system has no solution. If the other equation in the system is graphed in the *xy*-plane, what is the slope of the graph?

$-27r/7$
$-27/(7r)$
$7r/27$
$7/(27r)$

4.

Kenny engages in up to 3 types of exercise each week for a total of 6 hours while training for a triathlon. Kenny swims for the same number of minutes each week. The equation y = 360 − x − 170 represents the situation where Kenny bikes for *x* minutes during a week and runs for any remaining training time *y*, in minutes. If this equation is graphed in the *xy*-plane, which statement is the best interpretation of the *x*-intercept of the graph?

During a week when Kenny doesn't run, he bikes for 190 minutes.
During a week when Kenny doesn't bike, he runs for 190 minutes.
Each week, Kenny bikes and runs for a total of 190 minutes.
During a week when Kenny swims for 190 minutes, he bikes for 170 minutes.

5.

In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to 7x + 4y > 12?

The region where x > 0 and y > 0
The region where x < 0 and y > 0
The region where x < 0 and y < 0
The region where x > 0 and y < 0

6.

If (9x/2) − 11 = (45/2) − 11, what is the value of 9x?

5
10
45
54

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7.

$$z(w) = (0.829)^{2w}$$

The function z is defined by the equation shown. The value of $z(w)$ decreases by p% for each increase by 1 in the value of w. Which of the following is closest to the value of p?

- 0.171
- 0.829
- 17.1
- 31.3

8.

$$y = -4x^2 - 47$$
$$y = qx - 43$$

In the given system of equations, q is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

- −90
- −51
- −8
- −1

9.

For a certain circuit, its power P, in watts; current C, in amperes; voltage V, in volts; and resistance R, in ohms, are related as $CV^2 \div P = \sqrt{(PR)}$, where P, C, V, and R are positive. When $R = 18$, which equation correctly expresses P in terms of C and V?

- $P = CV^2 \div \sqrt{(18P)}$
- $P = CV^2 \div \sqrt{18}$
- $P = \sqrt[3]{(C^2V^4 \div 18)}$
- $P = \sqrt[3]{(18 \div (C^2V^4))}$

10.

A company determined that for a selling price of $120.00, zero products would be sold. For each $1.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly $1,966.50, which of the following could be the number of products sold? (revenue = price × number of products sold)

- 23
- 40
- 1,231
- 2,400

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11.

f(x) = √(6x + 7)

The function f is defined by the given equation. If f(a) = −9a, where a is a constant, what is the value of a?

−1/3
−7/27
7/27
1/3

12.

The positive number a is 2,106% of the sum of the positive numbers b and c, and b is 81% of c. What percent of b is a?

4,706%
2,600%
38.12%
21.87%

13.

The dot plots represent the distributions of values in data sets A and B.



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Which of the following statements must be true?

I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

I and II
I only
II only
Neither I nor II

14.

A right square pyramid has a total surface area of 28,160 square inches, and the combined surface area of the four lateral faces of this pyramid is 16,060 square inches. What is the height, in inches, of this pyramid?

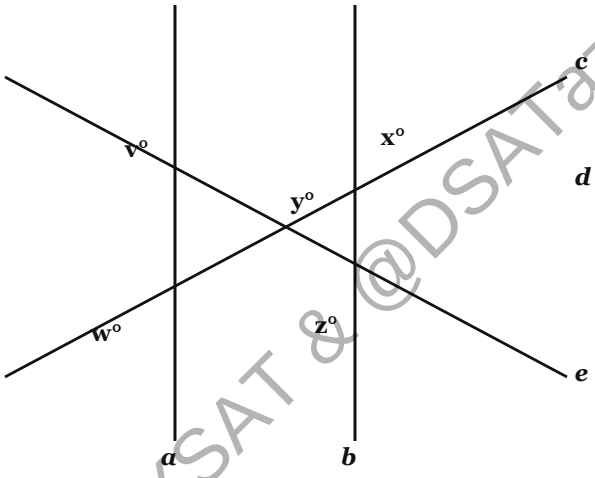
48

55

73

110

15.



Note: Figure not drawn to scale.

In the figure, parallel lines *a* and *b* are intersected by lines c, *d*, and *e*. If $z = 77$, $y = 108$, and $v < z$, which statement about x and w must be true?

x < w

x > w

x = w

x + w = 90

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16.

A linear model estimates the population of a city from 1995 to 2019. The model estimates the population was 56 thousand in 1995, 215 thousand in 2015, and x thousand in 2019. To the nearest whole number, what is the value of x?

Student-produced response — enter your answer in the grid.

17.

Jackson mows lawns in the summer. When Jackson mows a lawn with a fence, he charges $1.03 per minute. When Jackson mows a lawn without a fence, he charges $0.56 per minute. Jackson spends a total of 15 hours mowing lawns with and without a fence in a week and charges a total of $701.40 for mowing lawns during that week. How many minutes does Jackson spend mowing lawns with a fence during that week?

Student-produced response — enter your answer in the grid.

18.

Function f is a quadratic function where f(−20) = 0 and f(−4) = 0. The graph of y = f(x) in the xy-plane has a vertex at (r, −64). What is the value of r?

Student-produced response — enter your answer in the grid.

19.

The function p is defined by $p(x) = a((x + 5)^2 - b)((x + 5)^2 - c)$, where a, b, and c are constants. In the xy-plane, the graph of $y = p(x)$ passes through the points (−6, 30) and (0, 342). What is the value of $p(-10) + p(-4)$?

Student-produced response — enter your answer in the grid.

20.

The function f is defined by f(x) = |x|/a − 16, where a < 0. What is the product of f(17a) and f(8a)?

Student-produced response — enter your answer in the grid.

21.

As the size of a firework's shell increases, the height above the ground when the firework bursts increases. For a firework to burst 1,075 feet above the ground, a firework's shell in the shape of a sphere with a volume of 514.23 cubic inches could be used. To the nearest whole number, what is the volume of this firework's shell in cubic centimeters? (1 inch = 2.54 centimeters)

Student-produced response — enter your answer in the grid.

22.

In right triangle ABC, side BC has a length of √799. The hypotenuse, side AB, has a length of 32. What is the value of sin B?

Student-produced response — enter your answer in the grid.

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